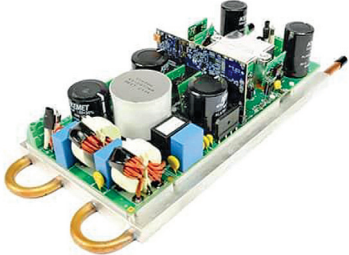


EV CHARGERS

The author, Nidhi Agarwal, works as Technology Journalist at EFY

6.6kW Bi-Directional EV On-Board Charger	7kW On-Board Charger	7.4kW On-Board Charger
		
The reference design is for a 6.6kW high power density bi-directional EV on-board charger. It provides an isolated adjustable DC output. This is achieved by using a single-phase AC input. This design is suitable for battery charging.	The reference design for an OBC has a 7kW unit capable of consistently delivering a current or voltage output. The standout feature of this design lies in its efficiency and high power density.	The reference design outlines a high-efficiency 7.4kW bidirectional, on-board charger, a crucial component for modern electric vehicles. Its versatility is evident in high-voltage battery power conversion systems, significantly optimising performance.
The CRD-06600FF065N-K, a reference design from Wolfspeed, is a 6.6kW high power density bi-directional EV on-board charger. The OBC operates on a universal, single-phase input, producing an output voltage ranging from 250V DC to 450V DC. It can operate in constant-current, constant-voltage, or constant power mode, adjustable through a controller area network (CAN) interface. The initial AC/DC power factor correction stage uses a continuous conductivity mode totem-pole bi-directional design at 67kHz, culminating in a DC link voltage. This voltage undergoes processing by a bi-directional isolated DC/DC CLLC resonant converter that operates between 148kHz and 300kHz. The design offers peak efficiency of 96.5% in both charging and discharging phases. Both input/output segments of the DC/DC stage employ a full bridge switch system separated by a high-frequency transformer. A specially designed heatsink manages the heat dissipation for all power MOSFETs and power magnetics. Though designed for EVs, it doesn't have an inbuilt battery charging algorithm. During discharging (DC/AC), a resistive load is crucial for testing, while both electronic and resistive loads serve the charging (AC/DC) phase, with an 18A maximum output current in charging mode.	The STDES-7KWOBK from STMicroelectronics is a reference design for an OBC that facilitates the charging of EV batteries. The design comprises two primary segments. The first involves an interleaved totem pole power factor correction (PFC) stage using a 2-channel interleaved topology, operating at 70kHz. The second features a dual galvanically-isolated full bridge LLC DC-DC ZVS resonant converter, set at 140kHz, employing MDmesh DM6 super-junction power MOSFETs. This reference design stands out with its 7kW unit that can consistently deliver current or voltage output, capable of operating individually or in a 3-phase mode, potentially amplifying its output to 21kW. The design's efficiency and high power density are underscored by the integration of silicon carbide (SiC), super-junction (SJ) power MOSFETs, silicon and SiC diodes, gate drivers, an SPC58NN84E7RMHBR automotive-grade microcontroller, and SCR thyristors that limit inrush current. Importantly, a 12V input supply voltage remains isolated from the output voltage ground connected to the high-voltage battery, and the design supports both constant current and voltage modes.	The TIDM-02013, a reference design from Texas Instruments (TI), employs a two-phase totem pole power factor correction (PFC) and a full-bridge CLLC converter with active synchronous rectification, optimising electrical conversion efficiency. This converter uses frequency and phase modulation techniques for output stability across a wide voltage range. A single TMS320F280039C microcontroller centrally controls the system, overseeing both PFC and DC-DC conversion stages for smooth operation. The design's inclusion of high-speed gallium nitride (GaN) switches, particularly the LMG3522-Q1 model, enhances its power density, achieving a peak efficiency of 96.5% and an open-frame power density of 3.8kW per litre. This showcases the design's capability to blend power and efficiency in a compact form. It operates best within a 90V to 264V alternating current voltage, typically stabilising at 240V AC, with primary and secondary voltages set at 400V DC and a 250V to 450V DC range, respectively. A standout feature is the CLLC resonant tank, operating with a 500kHz PWM switching capacity, promoting a higher power density. Additionally, it employs advanced soft switching, leveraging zero voltage switching (ZVS) on the primary side and blending ZVS with zero current switching (ZCS) on the secondary for superior efficiency.
The reference design is suitable for isolated bi-directional EV charging systems.	The reference design lets users power their EVs using household AC mains plug or private and public AC charging stations.	The reference design is a robust solution for DC fast charging stations, facilitating quicker, more efficient power conversions.
Wolfspeed	STMicroelectronics	Texas Instruments (TI)
https://www.electronicsforu.com/electronics-projects/reference-design-for-6-6-kw-bi-directional-ev-on-board-charger	https://www.electronicsforu.com/electronics-projects/reference-design-for-7-kw-on-board-charger	https://www.electronicsforu.com/electronics-projects/reference-design-for-7-4-kw-on-board-charger